



Evidence of double-funneling effect of beech trees by visualization of flow pathways using dye tracer

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SUMMARY

The aim of this study was to determine how the crown and root architecture of beech trees control stemflow formation, stemflow infiltration, and the subsurface transport of stemflow through the soil with a novel dye tracer experiment. With this approach, it is possible to apply the dye solution directly to the stem, enabling a more advanced technique to represent the flow processes than has been used in previous studies. Stemflow concentrates and channels water over localized circular areas at the tree base, as shown in this study. Tracing stemflow using dye demonstrated that infiltrating stemflow water was preferentially funneled belowground along heart and sinker roots, laterally oriented coarse roots, and congregations of fine roots grown into decayed root channels of felled trees. The results presented in this study reveal that stemflow and root induced preferential flow triggers fast lateral subsurface flow bypassing large parts of the soil matrix at least during rainstorm events. Based on our measurements, we hypothesize that the formation of stemflow and the channeling of infiltrating stemflow water along the roots of beech trees might be an important mechanism of runoff generation in forested catchments.

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1. Introduction

The conversion of even-aged Norway spruce (*Picea abies*) or Scots pine (*Pinus sylvestris*) monocultures into more structured and mixed forest stands is the main silvicultural goal in Germany and other European countries (e.g. Kint et al., 2006; De Schrijver et al., 2007; Berger et al., 2009; Heinrichs and Schmidt, 2009). The aim of this ongoing forest transformation is to reduce vulnerabilities due to insect outbreaks, climatic extremes (droughts, storms), and future climate change (e.g. Gärtner and Reif, 2004; Seidl et al., 2008). In this context, European beech (*Fagus sylvatica*) plays an important role when coniferous monocultures are converted into mixed forests by reintroduction of broadleaf tree species (Geßler et al., 2007; Poljanec et al., 2010). For instance, the German state Baden-Württemberg intends to increase the proportion of beech from the current 20% to more than 30% within this century (Moosmann, 2002 cit. in Geßler et al., 2007). As a more specific example, from the forest county Eibenstock (W. Ore mountains, Saxony) the public enterprise, Sachsenforst, plans to convert 40% (ca. 7900 ha) of the state owned spruce monocultures into mixed spruce-beech forest (Sonnemann and Baumann, 2011).

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Based on comparative studies to the water budgets between spruce and beech stands (e.g. Tuzinsky, 2000; Schume et al., 2004; De Schrijver et al., 2007; Schwärzel et al., 2009a), it can be expected that the reintroduction of European beech into former coniferous monocultures will result in a different relative contribution of evaporation, transpiration, and interception to total evapotranspiration. As a consequence, such shifts in evapotranspiration will also affect water yield and runoff at the catchment scale. Moreover, in contrast to Norway spruce and Scots pine, stemflow is a significant component of the water balance of European beech stands. As an example, analysis of precipitation events in an old beech forest have shown that events with precipitation >2 mm triggered stemflow, which then amounted to 16% of the annual rainfall (Schwärzel et al., 2009a).

Stemflow is the portion of precipitation intercepted by the canopy that flows down the trunk. In European beech stands, the generation of stemflow is due to a morphological peculiarity, namely a funnel-shaped crown – caused by steeply inclined branches and twigs – in combination with smooth bark. Stemflow concentrates and channels water over localized circular areas at the tree base (e.g. Crabtree and Trudgill, 1985; Durocher, 1990; Tanaka et al., 1996; Liang et al., 2007). Therefore, it was concluded that the infiltration of stemflow fluxes into the soil should be considered as a point source in the recharge process. In this context, Johnson and Lehmann (2006) introduced the term ‘double-funneling of trees’ because stemflow water entering the soil may be funneled

preferentially belowground along tree roots and macropores, bypassing much of the bulk soil. Although stemflow infiltration occurs at the pedon scale ($<1 \text{ m}^2$), triggered preferential flow processes may have a significant impact on the runoff generation at the catchment scale, at least during rainstorms (Crabtree and Trudgill, 1985; Levia and Frost, 2003; Liang et al., 2007). Thus, and in light of the ongoing forest transformation, redistribution of stemflow water under European beech needs to be investigated in more detail.

The most popular methods to study double-funneling processes of trees are (i) monitoring of soil water dynamics using tensiometers and soil moisture sensors (e.g. Durocher, 1990; Liang et al., 2007), (ii) monitoring of soil solute fluxes using suction cups (e.g. Crabtree and Trudgill, 1985; Chang and Matzner, 2000), and (iii) visualization of flow pathways using dye tracers (e.g. Martinez-Meza and Whitford, 1996; Li et al., 2009; Liang et al., 2009). Monitoring soil water and solute fluxes enables the quantification of flow rates under different boundary conditions over longer periods. In addition, explanations of ongoing transport processes underneath the trees can be inferred from such data (cf. Nieber and Sidle, 2010). In contrast to monitoring of water and solute fluxes, the use of a dye tracer enables the visualization of flow paths and flow networks. Due to destructive sampling, not only the dye-stained areas but also features (e.g. soil structure, soil texture, biopores, roots, stones, etc.) that may have an impact on soil water movement can be mapped (e.g. Anderson et al., 2009; Graham et al., 2010). Thus, underlying principles between soil properties and pathways for preferential soil water movement may be analyzed and generalized.

Only a few studies have traced root channelization of stemflow water with dyes. In these studies, dye powder was sprinkled on the soil surface around the base of trunks of desert shrubs (Martinez-Meza and Whitford, 1996; Li et al., 2009) or dye solution was sprayed toward the downslope side of tree stems of *Stewartia* plants (Liang et al., 2011). We designed a novel dye tracer experiment for tracking root channelization of stemflow water. The idea behind the novel design is that the dye solution is applied directly to the stem. With this approach, we assure natural flow processes along the trunk and coarse roots. Furthermore, our experimental setup enables the study of root channelization of stemflow water under controlled boundary conditions. The aim of our study was to examine to what extent the reintroduction of European beech in monocultures of Norway spruce stands might modify soil water pathways and subsurface flow processes. In addition, investigations to the anisotropy of hydraulic conductivity were carried out as a basis for the interpretation of the dye-traced flow processes.

2. Materials and methods

2.1. Study site

The study site (Landberg Station $50^{\circ}59'N$, $13^{\circ}29'E$, 400 m a.s.l.) is located in the Tharandt Forest $\sim 20 \text{ km}$ SW of Dresden (lower altitudes of the E. Ore Mountains, Saxony, Germany) on a plateau. The Tharandt Forest is characterized by a suboceanic climate. According to the long-term records of the Grillenburg weather station (1960–2007) the mean annual air temperature is 7.4°C and the mean annual precipitation is 879 mm. Landberg Station was founded in 2005, when a flux tower over a European beech stand and plots for soil hydrological measurement were installed (for details see Schwärzel et al., 2009b). The beech stand was established by wild juvenescence between 1908 and 1933. It is dominated by *F. sylvatica* (80.3%) along with 8.5% *P. abies*, 5.6% *P. sylvestris*, 0.5% *Betula pendula*, and 0.5% *Quercus spec.* The bedrock geology is highly variable (gneiss, rhyolite, sandstones and claystones from the Lower Cretaceous, basalt), forming a mosaic of various soil types (according to the World Reference Base WRB, 2006 mostly Dystric

Table 1
Field profile description of the test site soil. The soil is a Stagnic Luvisol (acc. to the World Reference Soil Base, 2006). Roman numerals before the soil horizon letter indicate an abrupt change in the parent material. Horizon symbols are designed according to the Ad-Hoc-Arbeitsgruppe Boden (2005). Diameter of the fine roots: $<2 \text{ mm}$.

Horizon symbol	Horizon characteristics	Thickness (dm)	Sand (M.%)	Silt (M.%)	Clay (M.%)	Soil structure	Fine root intensity (roots per dm^2)
Aeh	Weakly bleached by podsolization	7	15	67	18	Subpolyhedral	21–50
Sw-Al	Eluviated by clay migration and mottled due to perched water (features occupying $< 10\%$ area of the horizon)	19	11	72	17	Subpolyhedral	>50
Bt-Sdw	Enriched with clay due to illuviation and mottled due to perched water (features occupying between 10% and 50% area of the horizon)	22	11	63	26	Subpolyhedral	6–10
IIBv-Swd	Brunified and mottled due to perched water (features occupying between 10% and 50% area of the horizon)	54	11	71	18	Subpolyhedral	3–5
IIIBjSd	Ferriallitic (relic) and mottled due to perched water (features occupying between 10% and 50% area of the horizon)	22	50	23	27	Platy-like	1–2



Fig. 1. Setup of the stemflow experiment.

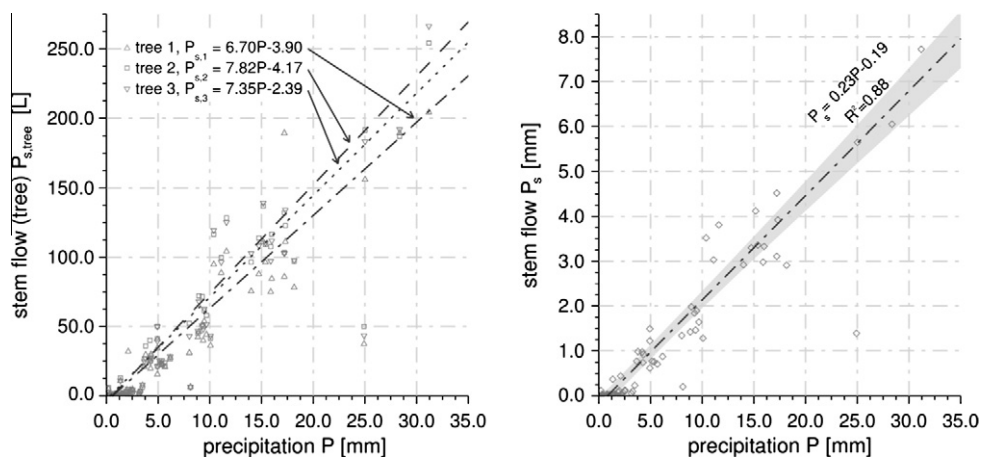


Fig. 2. Relationship between open-land precipitation and stemflow, measured at the test site (Landberg station, data from Spank, 2010) during the May-to-September period in 2006, 2007, and 2008. The diameter of breast height of the sampled beech trees are 0.335 m (tree 1), 0.413 m (tree 2), 0.385 m (tree 3). The projected canopy area of these trees are 18.68 m² (tree 1), 22.53 m² (tree 2), and 28.20 m² (tree 3).

Cambisols, Podisols, and Stagnosols) (see Schwärzel et al., 2009b). Geomorphologically, level and gentle-sloped plateaus (<2°) are predominant. Soils have mostly formed from Pleistocene periglacial cover deposits consisting of debris from the above mentioned parent materials and loess. The dye tracer experiment was conducted on a soil classified as Pseudogley–Parabraunerde according to the German soil classification (Ad-hoc-Arbeitsgruppe Boden, 2005), and as Stagnic Luvisol according to the WRB (2006). The investigated soil plot is situated on a gentle-sloped plateau (ca. 3% sloped). A soil profile description is shown in Table 1. The upper part of the profile is influenced by loess. In contrast to the topsoil, subsoil horizons are compacted and characterized by mottles and related redoximorphic features. These compacted layers were formed by periglacial solifluction prior to the soil formation and by illuviation of clay due to pedogenetic processes. As a result of soil compaction, stagnant soil water formation in deeper soil horizons is promoted. Therefore, the root intensity decreases sharply with soil depth (Table 1). Most of the fine roots were observed in the upper 40 cm of the soil profile. While the top- and subsoil horizons are characterized by a subpolyhedral soil structure, the horizon at the bottom of the soil profile shows a platy-like soil structure.

2.2. Measurements

2.2.1. Stem flow experiment using a dye tracer

For the stemflow tracer experiment, a linear sprinkler hose (GARDENA Sprinkler hose) was wound around a beech tree at breast height using a clamp arrangement (Fig. 1a). The linear

sprinkler hose is perforated on only one side. Therefore, the hose is well-suited to give a fine spray for narrow areas. The hose was connected via an electropump to a water tank. A valve was installed upstream of the pump to enable various flow rates. Depending on the valve settings, flow rates between 0.3 and 5 l min⁻¹ can be achieved. To assure a uniform wetting of the stem and to prevent splashing of water, a geotextile was placed between stem and hose.

On 10 November 2009, the stemflow experiment with the dye tracer was conducted. Before the experiment, soil water contents of the topsoil horizons were manually measured at our test site using TDR-technique. Soil water content was close to the water content at field capacity. The diameter at breast height of the sampled beech tree was 0.32 m and the projected canopy area was 17.2 m². Preliminary tests at the site have shown that a concentration of 4 g Brilliant Blue dye FCF per liter of water is needed to create colored flow paths that prominently stand out from the background. The characteristics of the applied dye tracer are given in Flury and Flühler (1995). Before the experiment, forest litter (thickness of the litter ca. 0.5 cm, the litter was not water-repellent) around the sample tree was removed and the soil surface was wetted with a spray of water. This was done to enable a better observation of the water entry into the soil, and if necessary (when water spread over the soil surface), to reduce the irrigation rate. The irrigation scheme for the stemflow experiment was designed based on stemflow and open-land precipitation data, measured at the Landberg station and a nearby weather station during the May-to-September period in 2006, 2007, and 2008 (Fig. 2).

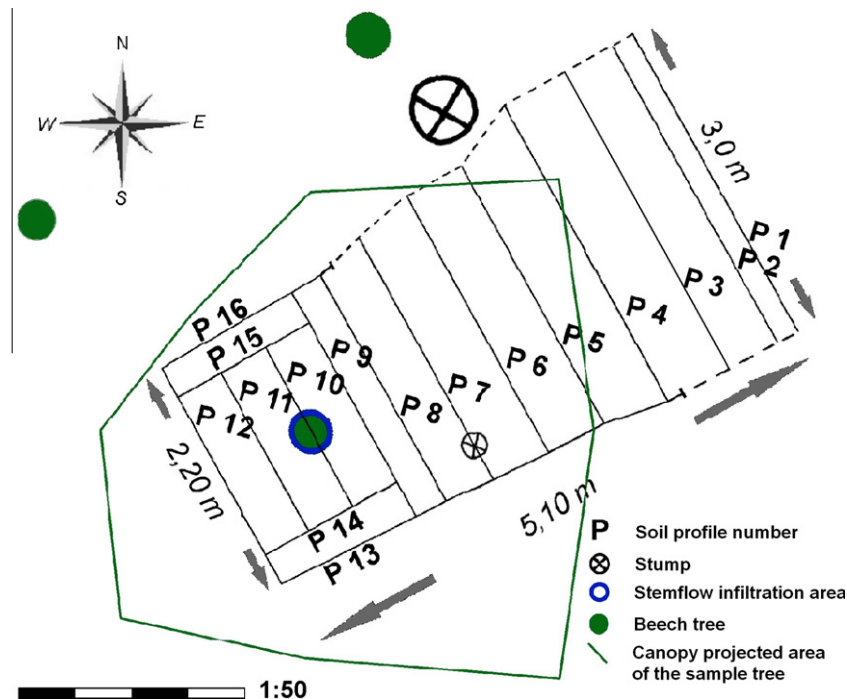


Fig. 3. Site for the stemflow experiment. In the scheme, the sample tree (green circle outlined in blue), the canopy projected area of the sample tree (green line), and the sampled soil profiles are included. These profiles were opened after the stemflow experiment. The diameter at breast height of the sampled beech tree was 0.32 m and the projected canopy area was 17.2 m². (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

During these years, rainstorm events lasted 4 h on average. In our dye tracer experiment, a rainstorm event of 18 mm was chosen for our stemflow experiment. According to the data in Fig. 2, such a rainstorm event creates a normalized stemflow of 4 mm (which corresponds to ca. 120 l of water that flows down the trunk). We applied such a stemflow over a period of 180 min. In total, 120 l of stemflow was applied, whereby the flow rate was varied from 0.3 to 2.5 l min⁻¹ during the experiment. At the end of the experiment, we cut down the sample tree and photographed the infiltration area of stemflow around the tree. This was later used to determine the dye colored area by tracing the contour of the blue-stained with Corel Photo-Paint. One day after the dye application, we opened a trench with a 1.0 m depth and 3.0 m width. This trench (profile 1 according to Fig. 3) was situated downhill from the stemflow infiltration area, 4.30 m away from the sample tree.

Blue-stained areas, coarse roots, dead roots, macropores, and soil horizon boundaries were transferred onto a plastic sheet. Over the next 9 days, a total of 16 soil profiles were prepared in this way (cf. Fig. 3). The transferred features were digitized and edited using Corel Photo-Paint, Corel Inc. Ottawa. The images were scaled to make one pixel equivalent to one square millimeter as in Weiler and Flüher (2004). Finally, areas of dye coverage and coarse roots per horizon were estimated by image processing similar to the study of Hangen et al. (2004) or Wessolek et al. (2007). In addition, the dye tracer distribution curve within the soil profile was derived. In addition, the dye tracer distribution curve within the soil profile was derived.

2.2.2. Soil hydraulic properties

Undisturbed soil cores in both vertical and horizontal directions (250 cm³ each direction with threefold repetition) from every horizon (with the exception of the topsoil horizon) were taken for laboratory measurements of bulk density, water retention curve, as well as saturated and unsaturated hydraulic conductivity. We simultaneously determined the unsaturated hydraulic conductivity and the water retention curve using the evaporation method

according to Wind (1968). Furthermore, saturated and near-saturated hydraulic conductivity of the topsoil horizon was determined using the hood infiltrometer (Schwärzel and Punzel, 2007). We measured the infiltration rate with pressure supply heads of 0, -1, and -2 cm at seven locations of the site. Before the tests, we removed the forest litter. Analysis of the measurements was based on Wooding's (1968) solution for infiltration from a circular source with a constant pressure head at the soil surface. Following all experiments, seven undisturbed soil cores (100 cm³) were extracted from beneath where the infiltration had been measured. The cores were used to determine the bulk densities.

2.3. Velocity calculations

As in the study of Anderson et al. (2009), we assumed that the total area of the dye-stained region for each cross-section corresponds to the cross sectional area of flow. Therefore, the Darcy velocity (V in [L T⁻¹]) can be calculated for each cross-section as follows:

$$V = Q/A \quad (1)$$

where Q is the applied irrigation rate [L³ T⁻¹] and A [L²] is the dye-stained area for each cross-section.

3. Results and discussion

3.1. Soil hydraulic properties

For a better understanding of the below presented results of the dye-traced flow patterns, a compilation of the soil hydraulic data is given in Table 2. We observed an increase in bulk density with increasing soil depth. Horizons with high bulk densities are characterized by very low air capacities. This can be derived from the water retention data, listed in Table 2. The drainable pore volume (between pF 1.0 and pF 2.0) of the B-Horizons is less than 3 vol.%.

Table 2

Soil hydraulic properties of the test site soil. δB = bulk density. K_{SAT_Field} = saturated hydraulic conductivity measured in the field using the hood infiltrometer. K_{SAT_Lab} = saturated hydraulic conductivity measured in the lab using the constant-head method. Unsaturated hydraulic conductivity and water retention was simultaneously measured in the lab using the evaporation method. h = pressure head.

Horizon	Sample orientation	K_{SAT_Field} ($cm\ d^{-1}$)	δB ($g\ cm^{-3}$)	K_{SAT_Lab} ($cm\ d^{-1}$)	Unsaturated hydraulic conductivity measured at $\log[h]$ in ($cm\ d^{-1}$)					Water retention measured at $\log[h]$ in (vol.%)				
					1.5	2.0	2.5	2.9		1.0	1.5	2.0	2.5	2.9
Aeh Sw-Al	Vertical	997	0.700	Not measured	1.26	.300	.025	.002		Not measured	56.6	49.5	34.9	24.8
	Horizontal	Not measured	1.104	698	.620	.164	.030	.003		61.0	54.2	46.8	34.1	24.4
Bt-Sdw	Vertical		1.165	621	.205	.084	.027	.009		60.6	40.8	39.4	36.2	32.6
	Horizontal		1.567	2.67	.151	.086	.034	.010		41.2	39.8	39.1	36.1	30.9
IIbV-Swd	Vertical		1.582	52.5	.171	.083	.037	.014		41.1	43.4	41.9	38.3	31.3
	Horizontal		1.524	5.71	.098	.034	.013	.004		43.9	40.8	39.7	37.1	33.6
IIIBjSd	Vertical		1.560	3.86	.111	.105	.013	.003		41.3	41.8	40.1	37.6	33.3
	Horizontal		1.535	2.02	.415	.390	.025	.006		42.2	42.0	40.6	37.3	33.1

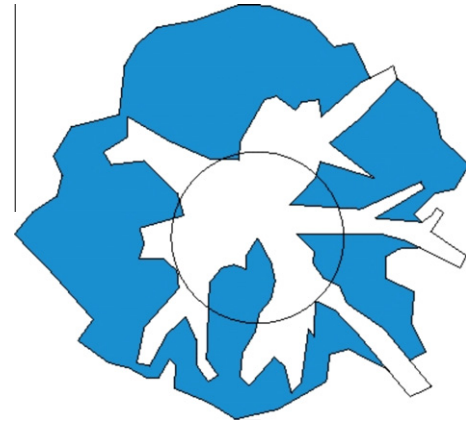


Fig. 4. The colored flow area around the sample tree. This area corresponds to a size of 0.245 m². Brilliant Blue was used as a dye tracer to visualize the portion of water that reaches the soil by flowing down the stem. The inner circle of the drawing is the diameter of the sample tree at breast height (0.32 m). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

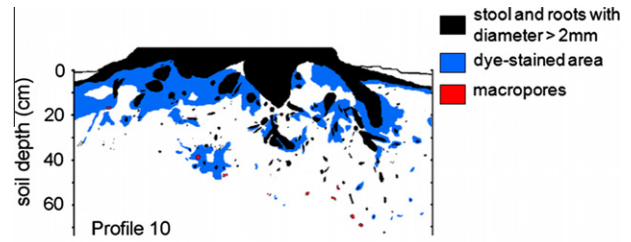


Fig. 5. Dye-stained areas under the stool of the sample tree. The width of the profile is ca. 120 cm. The macropores ($\varnothing > 2\ mm$), presented in this figure, originated from decayed roots and wormholes.

This finding is well-supported by the observed fine root patterns (cf. Table 1). Low air capacities and soil compaction of the subsoil horizons restricted fine root penetration. In contrast to the B-horizons, the A-horizons are well aerated (the drainable pore volume is >12 vol.%) and extremely densely rooted. For all horizons, no significant difference in water retention data between horizontal and vertical samples was observed (Table 2). Therefore, it can be concluded that mechanical disturbances of the soil during sampling was minimal.

A sharp drop in saturated hydraulic conductivity was found between the top- and subsoil horizons. For instance, the vertically saturated conductivity of the Sw-Al horizon was 700 $cm\ day^{-1}$, whereas the corresponding conductivities of the underlying horizons amounts to values between 2 and 6 $cm\ day^{-1}$ (Table 2). In almost all cases, no statistically significant differences in the hydraulic conductivity (saturated as well as unsaturated) between vertical and horizontal samples of each horizon were found. In most cases, differences between the vertical and horizontal conductivity data can be attributed to small scale soil heterogeneity rather than to anisotropy. However, there is a big exception. A drastic difference between the vertically and horizontally saturated hydraulic conductivity was detected in the clay enriched Bt-Sdw-horizon. The horizontal conductivity was 20 times larger than the vertical conductivity. We attributed these differences to decayed, horizontally oriented beech roots of felled trees.

3.2. Infiltration pathways traced by the dye experiment

As mentioned above, 120 l of stemflow water was applied over a period of 180 min. No formation of surface runoff was observed

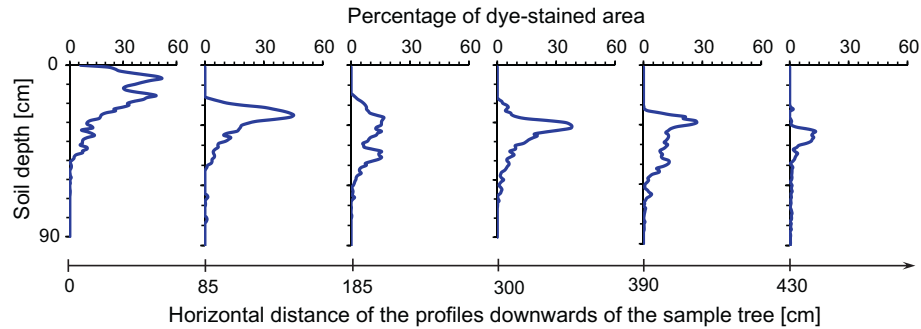


Fig. 6. Percentage of the dye-stained area with soil depth for various profiles situated under or downwards of the sample tree (cf. Fig. 3). There is a sharp drop in hydraulic conductivity between the topsoil and the subsoil horizons (the wavy boundary is approx. at 30 cm soil depth, cf. also Table 2).

throughout the whole experiment. Stemflow infiltrated in the immediate vicinity of the stem into the soil. It can be seen from Fig. 4 that the infiltration area around the tree is circular in shape which agrees well with the findings of Tanaka et al. (1996). This circular-shaped infiltration area has a diameter of 0.245 m². Therefore, the Darcy velocity amounts to 392 cm day⁻¹ (Eq. (1)), which is in the same magnitude as the measured saturated hydraulic conductivities of the topsoil horizons (cf. Table 2). Previous studies have indicated that the Brilliant Blue dye can be absorbed by the soil (cf. Flury and Flühler, 1995), resulting in a slight overestimation in the cross-sectional stained area. Thus, the calculated Darcy velocity may be underestimated.

Fig. 5 shows the redistribution of stemflow in the immediate vicinity of the trunk of the sample tree. The first 20 cm under the soil surface was almost entirely saturated by stemflow water. This indicates that water flow occurred primarily through the soil matrix of the topsoil horizons, with some stemflow water flowing also infiltrating along the root collar into the soil. Due to the high conductivity of the topsoil (cf. Table 2) water may have laterally moved from those flow paths (provided by the roots) into the soil matrix. It can be seen from Fig. 5 that to some extent, stemflow water preferentially penetrated along the heart roots of the beech and associated coarse roots deeper into the soil. In such situations, the soil around those roots was uniformly saturated by the dye.

In Fig. 6, the percentage of dye-stained area with soil depth is shown for six soil profiles situated downslope of the sample tree. As previously discussed, the infiltration of stemflow near the stem was predominately a homogeneous flow through the soil matrix of the topsoil horizons (Aeh and Sw-Al, cf. Table 1). For this reason, most of the dye-stained area can be found in the upper part of that cross-section (profile 10 in Fig. 6). Note that in the profiles situated downslope of the sample tree, the topsoil horizons were not wetted by the dye. With increasing distance from the stem, the dye-stained area was mainly found in the transition between the clay-depleted (Sw-Al, cf. Table 1) and clay-enriched horizons (Bt-Sdw, cf. Table 1) as well as in the clay-enriched horizon alone. In some cases (e.g. in a distance of 185 cm from the sample tree, Fig. 6), a noteworthy breakthrough of stemflow into the underlying Bv-Swd horizon was detected. Such a development was promoted by sinker roots and decayed sinker roots of felled trees. Furthermore, dye-stained areas were even observed 430 cm downslope from the stem (profile 1 in Fig. 6). Using Eq. (1) and the estimated cross-sectional dye-stained area of profile 1, we calculated a flow velocity of more than 2000 cm day⁻¹.

3.3. Interpretation of flow processes

From the tracer distribution curves presented in Fig. 6, the following flow processes can be deduced. Firstly, in the immediate vicinity of the stem, the infiltration of stemflow water into the soil

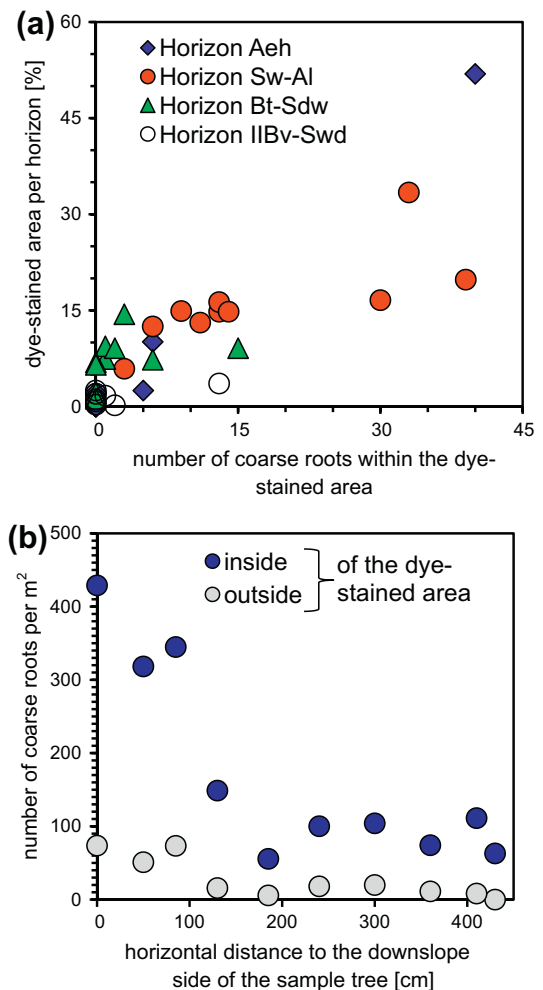


Fig. 7. (a) Relationship between the number of coarse roots within the dye-stained area per horizon and the percentage of dye-stained area per horizon (above) and (b) number of coarse roots inside and outside of the dye-stained area per profile. The diameter of coarse roots is ≥ 2 mm. For diagrams, data from the soil profiles situated under and downslope of the sample tree were used (soil profiles 1–10, cf. Fig. 3).

was mainly vertically directed. This attributed to the heart root system of the beech, which enables the entry of stemflow water into the soil, and to the high saturated hydraulic conductivities of the topsoil horizons, which enhanced the transport of stemflow water into the subsoil. The homogeneously blue-stained area in the direct vicinity of the stem indicates that the vertical transport of stemflow water through the topsoil horizons was mainly a uniform

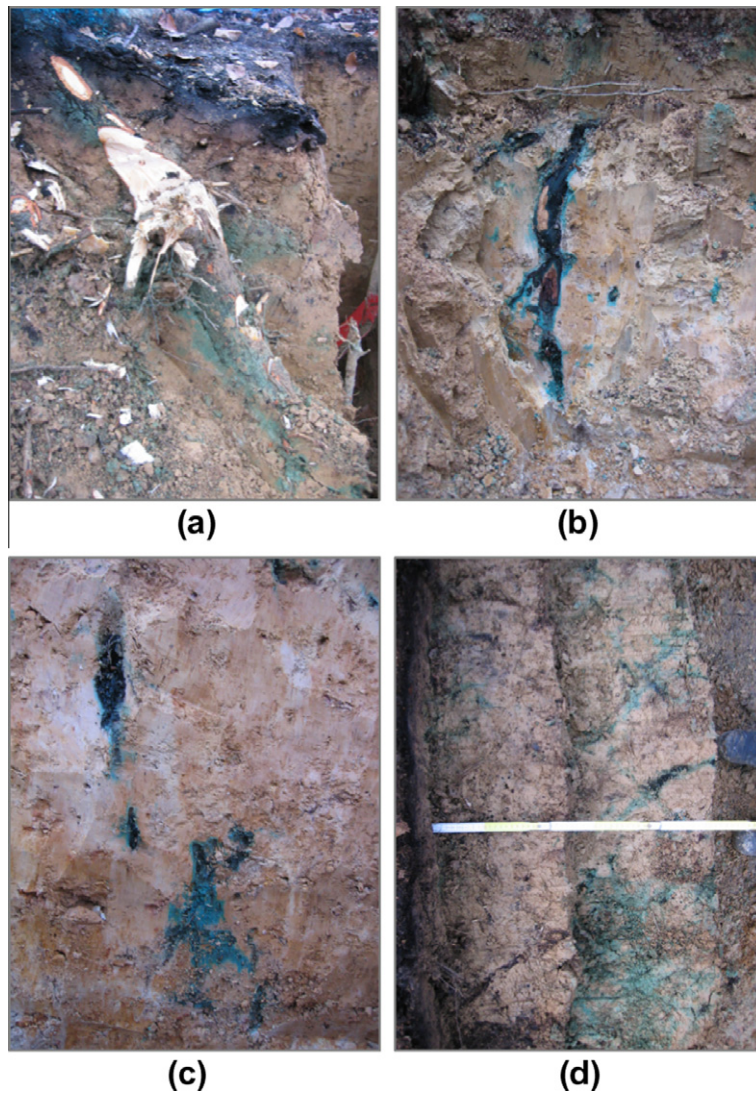


Fig. 8. The transport of stemflow water through the soil was visualized using Brilliant Blue. The photos show that stemflow water was preferentially channeled by (a) heart roots, (b) sinker roots, (c) large congregations of fine roots grown into decayed root channels of felled trees, and (d) fine roots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

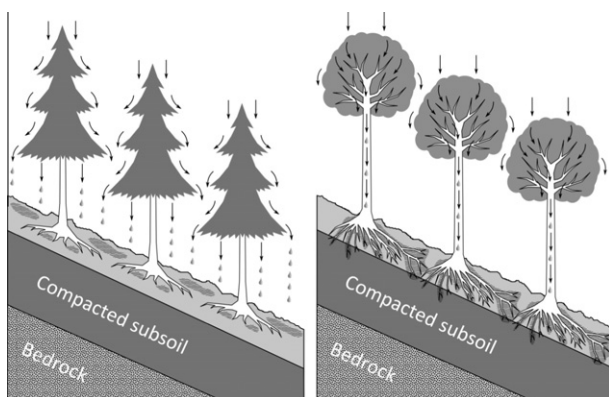


Fig. 9. Effect of crown and root architecture of spruce (left side) and beech (right side) on spatial distribution of precipitation within the stand and on transport of infiltrating precipitation through the soil.

matrix flow. To some extent stemflow water guided along the root collar into the soil may have laterally moved from those flow paths

into the soil matrix. Secondly, despite the gentle-sloped plateau, the vertically directed transport of stemflow water through the soil was redirected to the downslope side of the sample tree. What were the reasons for this flow phenomenon? On the one hand, a further vertical distribution of stemflow water was impeded due to a sharp drop in hydraulic conductivity of the subsoil horizons (cf. Table 2). Therefore, stemflow water perched at these horizons and moved laterally to the downslope side of the sample tree. On the other hand, visualization of flow patterns indicates that the lateral flow took place mainly in the transition zone between the clay-depleted (Sw-Al horizon) and clay-enriched (Bt-Sdw horizon) horizon as well as in the upper part of the clay-enriched horizon. There, laterally oriented coarse roots and decayed root channels of felled trees promoted the preferential flow of water, bypassing large parts of the soil matrix. Beech roots generally avoid growing in compacted soil horizons. This is likely due to a lack of oxygen (cf. Schwärzel et al., 2009a). Therefore, lateral spread of coarse beech roots is typical for such soil conditions as those prevailing at our test site (cf. Table 2). Finally, significant formation of deep seepage as a result of stemflow can be ruled out to the highly compacted subsoil.

The hydrologically significant role of the root system for the preferential routing of water through the soil clarifies (Fig. 7). Despite the scatter, the dye-stained area per horizon enlarged with the increasing numbers of coarse roots within the dye-stained areas (Fig. 7a). Moreover, the number of coarse roots in the blue-stained was eight times greater than average value for all profiles (Fig. 7b).

Additional evidence that the root network channeled stemflow water to the downslope side of the tree is presented in Fig. 8. Heart roots, sinker roots, fine root congregations grown into decayed root channels of felled trees, and fine roots served as the preferred pathways for the rapid transmission of stemflow water through the soil.

3.4. Double-funneling effect of trees as a mechanism of runoff generation

Based on a review of numerous studies on the fast subsurface flow in sloped terrain, it was suggested that pipe networks can control the hillslope's response to rainstorms (Weiler et al., 2006; Nieber and Sidle, 2010; Bachmair and Weiler, 2011). Dye staining experiments have shown that subsurface flow may often take place in living and decayed root channels (e.g. Noguchi et al., 1999; Anderson et al., 2009; Liang et al., 2011). This was confirmed by our study where lateral subsurface flow was promoted by the development of the root network. Visualization of the flow pathways and analysis of hydrologically relevant soil features has shown that the development of this root network depends on the soil conditions. The pronounced lateral spread of the root system of beech was caused by the compaction of the subsoil. As discussed earlier, subsoil compaction has formed by periglacial solifluction prior to the soil formation and by illuviation of clay due to pedogenetic processes. In contrast to beech, spruce trees generally develop a very shallow root system (for details see Schwärzel et al., 2009a), at least when the subsoil is compacted as in our study.

The formation of stemflow depends on both precipitation and tree characteristics (Levia and Frost, 2003). The schematic in Fig. 9 shows how the crown architecture of Norway spruce and European beech might control the spatial distribution of precipitation within these stands at the slope scale during rainstorms. Under spruce, precipitation input is rather evenly distributed whereas, under beech, precipitation is concentrated and channeled over localized circular areas at the tree base. Fig. 9 shows how the differences in the crown and root architecture between Norway spruce and European beech may result in different hydrologic responses to rainstorms. Formation of stemflow and induced stemflow infiltration in the immediate vicinity of the beech tree creates spatially disconnected, more or less, saturated soil patches (named as hydrologically active areas by Uchida et al. (2001)) along the slope. During rainstorms, these isolated hydrologically active areas are connected by laterally oriented living and decayed roots which were formed by the beech stand in response to soil compaction. Therefore, in contrast to spruce stands (assuming the same precipitation characteristics), we suggest that in beech stands the formation of lateral subsurface flow as a result of stemflow and root induced preferential flow might trigger a fast(er) hydrological response to rainstorms. Our observations and suggestions fit well to the recently proposed connect-and-react mechanism of Bachmair and Weiler (2011). They hypothesized that a significant response to rainstorm is triggered only if hydrologically active areas are interconnected.

4. Summary and conclusions

Due to the funnel-shaped beech crown in combination with smooth bark, beech trees produce a significant amount of stemflow during rainstorms. Our novel dye tracer experiment enables the study of how stemflow formation and infiltration effects soil water

redistribution processes. Stemflow water flows down the trunk and infiltrates into the direct vicinity of the trunk as presented in this study. It was shown that the root network controls the entry of stemflow water into the soil and the redistribution of stemflow water within the soil. At our test site, the vertical penetration of infiltrating stemflow water into the deeper soil was hampered by the sharp drop in hydraulic conductivity of the clay-enriched horizon. These low hydraulic conductivities have developed as a result of periglacial solifluction prior to the soil formation and by illuviation of clay due to pedogenetic processes. Due to the low conductivity of the subsoil, the vertically directed transport switched to a laterally directed downslope transport of stemflow water. The dye tracer experiment proved that the vertically directed flow took place mainly through the soil matrix, whereas the lateral redistribution of water took place along pathways provided by the root network. Due to channeling of stemflow water by laterally oriented roots, resp. decayed root channels, large parts of the soil matrix were bypassed and rapid interflow to the downslope side of the sample tree was observed.

Despite only a single tree was sampled we conclude that reintroduction of European beech in monocultures of Norway spruce stands will modify soil water pathways and subsurface flow processes. We hypothesize that old growths of beech trees situated on slopes with compacted subsoils develop a root network which promotes the preferential flow of stemflow water through the soil. In other words, double funneling (sensu Johnson and Lehmann (2006)) of beech trees might be an important mechanism in the generation of runoff in forested catchments. To test these hypotheses, stemflow-induced runoff formation needs to be studied at the hillslope scale. For it, our experimental setup could be installed at several trees along a slope. In addition, the hillslope should be equipped with piezometers and for the quantification of subsurface flow a setup might be chosen as proposed by Noguchi et al. (2001) or Ticehurst (2004). These new observation may lead to better conceptions of models taking into account stemflow-induced formation of runoff.

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